9 Databases

1 a Explain the meaning of the following database terminology.

Table: .....

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Record: .....

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Field: .....

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b Identify **four** basic data types used in a database and provide an example for each one.

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2 A single-table database, ICECREAM, has been set up to store the ice creams available for sale. The details included about each type of ice cream are:

- Type – for example, 'choc ice', 'lolly' or 'cone'
- Flavour – for example, 'raspberry'
- Size – 'Small', 'Medium' or 'Large'
- Number in stock – for example, '34'
- Re-order level – for example, '20'.

a Write down names for the five fields that would be required. For each field state with a reason the data type that should be used and give a sample of the contents of that field.

Field 1: .....

Data type: .....

Reason: .....

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Sample: .....

Field 2: .....

Data type: .....

Reason: .....

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Sample: .....

Field 3: .....

Data type: .....

Reason: .....

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Sample: .....

Field 4: .....

Data type: .....

Reason: .....

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Sample: .....

Field 5: .....

Data type: .....

Reason: .....

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Sample: .....

b i Explain why a primary key is needed in a database table.

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ii Give the reason why none of the existing fields are suitable to use as a primary key.

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iii Give a suitable field name, data type and sample for another field that could be used as a primary key.

Name: .....

Data type: .....

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Sample: .....

c Build the database table as specified in parts **a** and **b**. Write or paste a copy of your table structure here.

d Populate the database table with **six** records. Write or paste a copy of your populated table here.

3 a Explain what is meant by SQL.

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b Explain the function of each of the following SQL statements.

i SELECT

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ii FROM

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iii WHERE

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iv SUM

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c Identify **two** more SQL statements that you need to know for IGCSE Computer Science.

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d Use the six statements from parts **b** and **c** to write SQL queries for the database table you created in Question **2** to:

- Show the types, in alphabetical order, and sizes of all the ice creams that are in stock
- Show the total number of ice creams available to buy
- Count the number of ice creams that have stock below the reorder level.

Write or paste a copy of your SQL queries here.

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4 A single-table database, TEACHER, contains the details of the teachers in a school. The database includes these fields:

Name – the teacher's family name, for example, 'Yo'

Title – the teacher's title, for example, 'Mr'

Licence – the teacher's licence number, for example, 'L1234'

Gender – 'M' or 'F'

Subject – the main subject taught by the teacher, for example, 'Mathematics'

Class – the class the teacher tutors, for example, '2Y'

a i Identify the field that would be most suitable to use as a primary key. Give a reason for your choice.

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ii State the data type that could be used for each field.

Name: .....

Title: .....

Licence: .....

Gender: .....

Subject: .....

Class: .....

iii Build a database with the six records shown.

b i Identify a field that should be verified. Give a reason for your choice.

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ii Identify a field that could be validated. Give a reason for your choice.

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iii Add your validation to your database.

Here is part of the database table, TEACHER:

Name	Title	Licence	Gender	Subject	Class
Yo	Mr	L4579	M	Mathematics	3Z
Sing	Miss	L6713	F	Science	2X
Patel	Mr	L5421	M	Geography	4Y
Teo	Mrs	L7681	F	Mathematics	2Z
Young	Mrs	L6789	F	English	3X
Ling	Mr	L4980	M	Science	4X

c i State the output from this SQL query.

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SELECT Title, Name
FROM TEACHER
WHERE Subject = 'Science';
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ii Add the SQL statement to show the results in alphabetical order.

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iii Rewrite this SQL statement to include the mathematics teachers as well.

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iv Write a new SQL statement to count the number of mathematics teachers in the TEACHER table.

v Check that the SQL statements work for your database.

d Write an SQL statement to display the names of all the teachers and their classes in ascending order of class.

Check that your SQL statement works as expected.